Ai Assisted Coding

Name:U.Pranav Dev

Enroll.no:2503a52694

Batch:24

Responsible AI

Task Description #1 (Privacy&Security in File Handling)

Task: Use an AI tool to generate a Python script that stores user data (name, email, password) in a file.

Analyze: Check if the AI stores sensitive data in plain text or without encryption.

Input

```
import hashlib

name = input("Enter name: ")
email = input("Enter email: ")
password = input("Enter password: ")

hashed_password = hashlib.sha256(
    password.encode()
).hexdigest()

with open("user_data.txt", "w") as file:
    file.write(f"Name: {name}\n")
    file.write(f"Email: {email}\n")
    file.write(f"Password Hash: {hashed_password}\n")

print("User data saved securely.")
print("Password is stored as a hash.")
```

Output

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> python lab2_task1.py

Enter name: Lakshitha

Enter email: lakshitha@gmail.com

Enter password: abc123


User data saved securely.

Password is stored as a hash.
```

Task Description #2– Fair Decision Logic

Task: Prompt an AI tool to generate a scholarship eligibility checker based on academic score,

family income, and location. Examine whether the logic unfairly favors or disadvantages any

group.

Input

```python
academic_score = float(input("Enter academic score: "))
income = float(input("Enter family income: "))
location = input("Enter location: ")

eligible = (
    academic_score >= 75
    and income <= 300000
    and location.lower() in ["rural", "urban"]
)

if eligible:
    print("Scholarship Eligible")
else:
    print("Scholarship Not Eligible")

print("Decision is based on score and income.")
print("Location is not used to favor a group.")
```

Output

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> python lab2_task2.py
Enter academic score: 82
Enter family income: 250000
Enter location: rural

Scholarship Eligible
Decision is based on score and income.
Location is not used to favor a group.
```

Task Description #3 (Transparency in AI-Based Admission System)

Task: Use an AI tool to generate a Python program that predicts whether a student should be selected or rejected for admission based on factors such as entrance exam score, academic marks, and other relevant admission criteria.

Analyze: Check whether the AI system provides a clear and understandable reason for its admission

decision.

Input

```
entrance = float(input("Enter entrance exam score: "))
marks = float(input("Enter academic marks: "))

score = (entrance * 0.6) + (marks * 0.4)

if score >= 70:
    decision = "Selected"
    reason = "The combined score meets the admission threshold."
else:
    decision = "Rejected"
    reason = "The combined score is below the admission threshold."

print("Admission Decision:", decision)
print("Reason:", reason)
print("Combined Score:", round(score, 2))
```

Output

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> python lab2_task3.py

Enter entrance exam score: 78

Enter academic marks: 82


Admission Decision: Selected

Reason: The combined score meets the admission threshold.

Combined Score: 79.6
```

Task Description #4 – Ethical Scoring System

Task: Generate an employee performance evaluation system using inputs such as project completion rate, teamwork score, and

attendance. Analyze the scoring logic for any unethical weighting or hidden bias.

Input

```
completion = float(input("Project completion rate (%): "))
teamwork = float(input("Teamwork score (%): "))
attendance = float(input("Attendance (%): "))

performance = (
    completion * 0.50
    + teamwork * 0.30
    + attendance * 0.20
)

print("Performance Score:", round(performance, 2))

if performance >= 75:
    print("Evaluation: Good Performance")
else:
    print("Evaluation: Needs Improvement")

print("Weights are based on job-related criteria.")
```

Output

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> python lab2_task4.py
Project completion rate (%): 90
Teamwork score (%): 80
Attendance (%): 95

Performance Score: 87.0
Evaluation: Good Performance
Weights are based on job-related criteria.
```

Task Description #5– Accessibility and Inclusiveness

Task: Use an AI tool to generate a user feedback form application. Review whether the language and options provided are inclusive and accessible to diverse users.

Input

```
print("=== User Feedback Form ===")

name = input("Name (optional): ")
feedback = input("Please share your feedback: ")

print("\nHow was your experience?")
print("1. Very Good")
print("2. Good")
print("3. Neutral")
print("4. Needs Improvement")
rating = input("Choose an option (1-4): ")

if rating in ["1", "2", "3", "4"]:
    print("\nThank you for your feedback!")
    print("Your response has been recorded.")
else:
    print("\nPlease choose an option from 1 to 4.")

print("Name is optional and the form uses clear language.")
```

Output

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> python lab2_task5.py

=== User Feedback Form ===

Name (optional): Lakshitha

Please share your feedback: Easy to use


How was your experience?

1. Very Good

2. Good

3. Neutral

4. Needs Improvement

Choose an option (1-4): 1


Thank you for your feedback!

Your response has been recorded.

Name is optional and the form uses clear language.
```

Task Description #6 (Fairness in Loan Approval System)

Task: Use an AI tool to generate a Python program that predicts whether a person should be approved or rejected for a loan based on information such as income, age, employment status, credit score, and loan amount.

Analyze: Check whether the AI-generated system makes decisions unfairly based on gender, religion, race, or other sensitive/unrelated personal attributes.

Input

```
income = float(input("Enter annual income: "))
age = int(input("Enter age: "))
employment = input("Employment status: ")
credit_score = int(input("Enter credit score: "))
loan_amount = float(input("Enter loan amount: "))

approved = (
    income >= 300000
    and age >= 18
    and employment.lower() == "employed"
    and credit_score >= 650
    and loan_amount <= income * 0.5
)

if approved:
    print("Loan Decision: Approved")
    print("Reason: Financial criteria are satisfied.")
else:
    print("Loan Decision: Rejected")
    print("Reason: One or more financial criteria are not satisfied.")

print("No gender, religion, or race is used.")
```

Output

```
PS C:\Users\ASUS\OneDrive\Desktop\AAIC-LAB> python lab2_task6.py
Enter annual income: 600000
Enter age: 25
Employment status: employed
Enter credit score: 720
Enter loan amount: 200000

Loan Decision: Approved
Reason: Financial criteria are satisfied.
No gender, religion, or race is used.
```